

MACAU ANGLICAN COLLEGE



FISH LOGIC SIREN



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Abstract

Siren is Fish Logic's seventh Remotely Operated Vehicle (ROV), designed on the request for proposals by MATE Center and our wider global community for an ROV to assist in marine renewable energy, offshore aquaculture and monitoring the impacts of climate change. To meet the desired functionalities, Fish Logic, a company dedicated to developing underwater ROVs, has designed Siren with a flexible configuration. Siren is capable of installing offshore solar panels and underwater cameras, deploying BGC Floats to monitor ocean conditions and administering treatment to diseased corals.

Siren is fully designed by the twenty-one dedicated members of Fish Logic. The design is an evolution of Fish Logic's previous ROV, Otodus, with an emphasis on prioritizing safety and functionality, followed by ease of use, ease of maintenance, and ease of manufacture, Fish Logic has incorporated systems engineering in designing the ROV for better system integration with all aspects of the ROV validated through testing.

The majority of Siren's structure is 3D printed, allowing the structure to take on a very unconventional form that is modular and standardized. The structure supports six brushless thrusters and tools which can be "hot swapped". Otodus was designed to meet size and weight restrictions, with the resulting ROV being a high-performance, flexible, reliable, pilot-optimized and mission-oriented vehicle.

The development period of the Siren encompasses 8 months with about 4490 work hours. The market value of Siren is 32083.58HKD



Fish Logic Team Picture (photo by Ryan Chan)

Front Row (from left to right): Mr Andy, Yoanna, Isla, Onching, Hou Fong, Kimi, Kevin, Charles, Stephanie Alison, Daniel, Mr Ryan

Back Row: Stacy, Elli, Eric, Matthew, Tin Nam, Marcus, Wyatt, Aiden, Wells, Grace

Project Management

Fish Logic utilizes Agile project management for its focus on continuous incremental iteration throughout the ROV development. By utilizing Agile, the team aims to achieve a working product as soon as possible, which allows for immediate feedback throughout the development process and collaborative decision-making. Instead of strict targets, Agile establishes a broad vision at the start of the development process, allowing for flexibility and improvement during the development phases. When presented with changes in conditions, the team is given room to easily adapt. As the development of the ROV progresses, the team narrows down on the target incrementally with more clarity. Since Agile requires frequent feedback and collaborative decision-making, it is well suited to Fish Logic's small team and transparent work culture.

Scrum is the chosen agile project management methodology, in which the team works in short bursts from 2-4 weeks called sprints. Thus, the year development period is divided up and plotted onto a Release Planning Timeline. Its length is set so that each sprint preferably ends on an event, such as an experiment date, system test date, or water test date, and always requires deliverables such as a prototype part, tool, modifications, or software release to be completed.

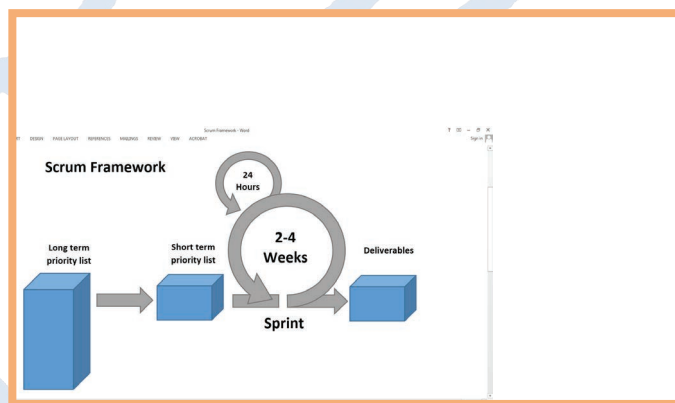


Fig. 1: Scrum Framework

To implement scrum, there are 4 types of documents known as artifacts: release planning, product backlog, sprint backlog, and burndown chart.

Product Backlog, which is a priority list of deliverables that covers the entire scope of the project that the team needs to achieve for each sprint, and the sprint backlog, a priority list developed at the start of each sprint of all the tasks to be completed in the current sprint. Previously, both lists were written

in Notability. However, other team members were not able to contribute to the list/documents and progress reports were slow as they needed to be manually created. Thus, Fish Logic has continued to use Jira, a proprietary agile management software that allows team members to view the current progress of the sprint, and the backlog of tasks yet to be completed before the sprint ends, as well as access to update the progress of their tasks. Jira also automatically generates a burn-down chart and cumulative flow diagram of all the deliverables, which shows the overall progress of the project. Our CEO, in this case, referred to as the Scrum Master selects tasks from the sprint backlog and briefs the team in a daily standup. Any obstacles to upcoming tasks are also discussed and dealt with. Scrum has allowed Fish Logic to remain adaptable as schedules are often modified due to unknown water testing dates, academic work of the team members, public exams, and unforeseen events.

After the end of each sprint, a sprint review is conducted to gather all data from water testing, including video footage, software bugs, and feedback from the pilot. This is used in a debrief meeting to analyze problems that need to be addressed, along with discussing what tasks have been completed and what work has still yet to be completed. A sprint retrospective is then conducted to inspect the working approach of the team, as well as look for improvements in our work method for the next sprint.

In regards to resources, it is easy to lose sight of resources allocated for the project as Scrum Agile management is less structured. However, by following the resource management principle of promoting the transparency of resources to other potential users in the team, the team can keep track of the

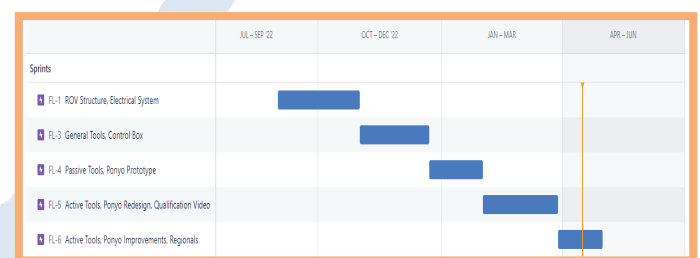


Fig. 2: Release Planning

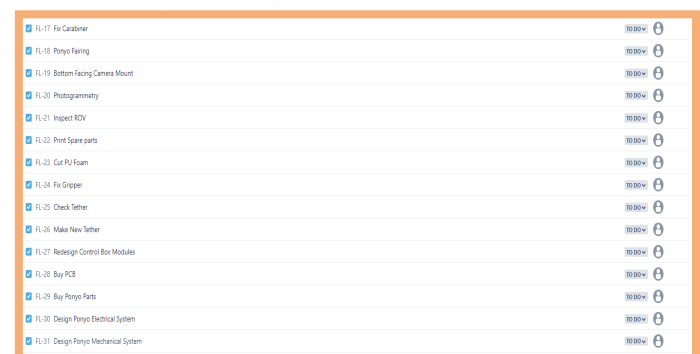


Fig. 3: Long Term Priority List

ID	Task Name	Status	Link
10-17	POVID - Backup Engine	ACTIVE TOOLS, PENDING REVISION...	SOME
10-18	POVID - Bladder Cage	ACTIVE TOOLS, PENDING REVISION...	IN PROGRESS
10-19	POVID - Battery Box	ACTIVE TOOLS, PENDING REVISION...	SOME
10-20	POVID - Electronics	ACTIVE TOOLS, PENDING REVISION...	SOME
10-21	POVID - Magnet Lining	ACTIVE TOOLS, PENDING REVISION...	SOME
10-22	POVID - Arduino Code	ACTIVE TOOLS, PENDING REVISION...	IN PROGRESS
10-23	TOOL - Spring Prototype	ACTIVE TOOLS, PENDING REVISION...	SOME
10-24	TOOL - UV Torch	ACTIVE TOOLS, PENDING REVISION...	IN PROGRESS
10-25	TOOL - Spring Arm	PENDING TOOLS, PENDING REVISION...	SOME

Fig. 4: Short Term Priority List

purchasing and allocation of materials. A spreadsheet is used to track resources of each sprint cycle with a special indication for resources that extend through more than one sprint cycle. Since Scrum uses rapid iterations in development, instead of committing to purchasing large amounts of resources as there may be changes in development direction, purchase decisions are executed only when initial test results justify committing resources needed for the next iteration in the purchase list.



Fig. 5: Burn Down Chart

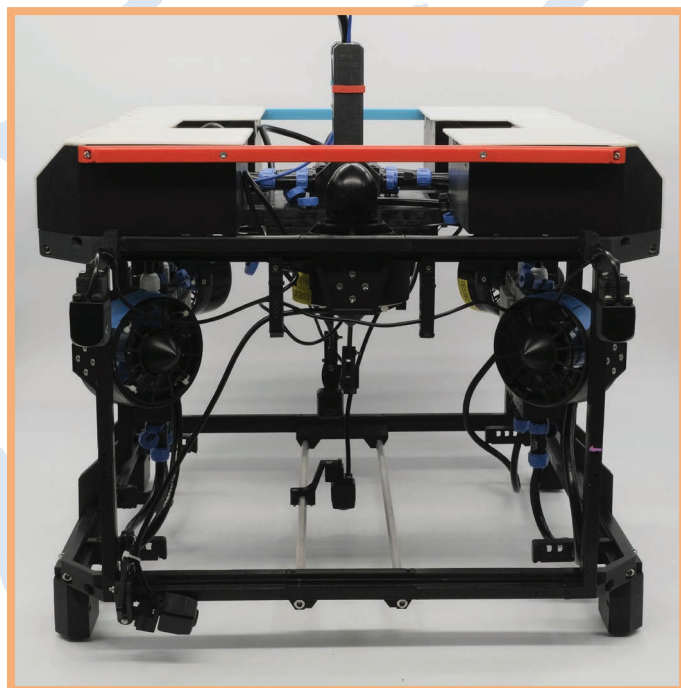


Fig. 6: ROV in real life

Design Rationale

Design Evolution

Fish Logic continues to use many of the features and design principles carried on from previous years, established ranging back from the ROVs: Leviathan (2017), Blazin' Hydra (2018), Electro Stargazer (2019), Hydron (2021) and Otodus (2022). The principles are as follows: Safety, Functionality, Pilot Oriented Design, Ease of Use, Ease of Maintenance, and Ease of Manufacture, with all aspects of the ROV validated through testing.

The safety of personnel, wildlife, and the environment is our number one priority in Fish Logic. Functionality is evaluated by the ability to perform all mission tasks in the allocated time. Fish Logic places heavy emphasis on pilot-oriented design. Good visibility, individualized control schemes, and custom set-ups based on pilot preference and feedback has shown to increase the speed, accuracy, and consistency of the pilot in completing the mission. Ease of use, which includes attention to ROV stability, predictability, and corrective thrust vectoring improves the confidence of the pilot. Ease of maintenance reduces the amount of downtime from swapping tools and allows for a quick return to water during operation. This is ensured by focusing on reliability, standardization, and modular parts. For mechanical reliability, simpler, passive design options are favored over active tools which are used when tasks require higher precision motion. The Standard Mounting System (SMS5) is used for the mounting of mechanical parts, allowing any tools and payload with SMS5 to be easily interchangeable. Front-to-back symmetry of the ROV allows the ROV to carry double the payload on both ends, reducing the need for surfacing and increasing efficiency. Electrical standardization is achieved with the Universal connectors and CAN bus. CAN bus is a bus network in which modules can be connected from anywhere in the network, allowing the ROV to have a flexible and scalable configuration. Interdependent electronics are housed in the same module to minimize the interface between modules and simplify maintenance. Electronic components are cast in epoxy to waterproof them reliably, eliminating all points of water entry. Most of the ROV parts were created using 3D printing. This not only guarantees a consistent quality, but also allows team members to focus on the designing of new parts whilst the current ones are being fabricated. Fish Logic relies on these design principles to eliminate the risk of technical debt build-up during the design phase. Upon completion, all parts will be validated using data driven scientific processes as well as feedback from both the pilot and the operation team.

Although passive tools are often preferred over active ones, by utilizing standardized and modular active components such as ESC hubs and motors designed from previous years of competing, the time taken to design and build new active tools, as well as the time taken to complete the task — without compromising on reliability — has been drastically reduced.

In order to take on a more comprehensive approach, Fish Logic decided to fully utilize systems engineering principles in designing the ROV for better system integration. With the ROV tool layouts and mission programs designed in conjunction with each other and according to pilot preferences, the approach to each task is optimized to minimize the effect on other tools.

Systems Engineering

Fish Logic uses systems engineering processes, inspired by SpaceX, due to its compatibility with Scrum Agile Project Management. Systems engineering is used to help develop safe, reliable and well-integrated systems in accordance with our design principles by anticipating and solving integration problems ahead of time. A balance between intensive pre-planned systems engineering and rapid prototyping is required to reduce system risk, with the balancing heavily dependent on the organizational agility, cost of iteration and the ability to trade lower level requirements. The team strives to learn through experience rather than consuming schedules attempting to anticipate all possible system interactions. To promote better systems integration, systems level tasks are distributed to departments to get each to focus on systems thinking. Which is then reviewed by the CTO who acts as the overall systems integrator. Only top level requirements are defined, tracked and verified but everything below these requirements is constantly traded and optimized during the design phase. This is to prevent derived requirements established in prior meetings from limiting the creativity of new ideas and solutions.

Fish Logic applies a systematic process for each tool during development. Once initial prototypes of tools enter the testing phase, the component interactions with the rest of the system are considered. Before tool development, the ROV cameras had their field of view and minimum focus distance measured. Once the distance is established, since the main camera is located in the center of the ROV, new tool placements and lengths are adjusted to the set distance. Tools that require precision will have dedicated secondary cameras provided for complimentary camera angles. Interactions with other tools are studied once tool placement and camera angles are set. With tools mainly placed at the front and back of the ROV, tools on each side are carefully configured to minimize

the interference between each other. Once all functional, pilot preferences and ease of use issues are addressed, and the tool design is optimized for manufacturing.

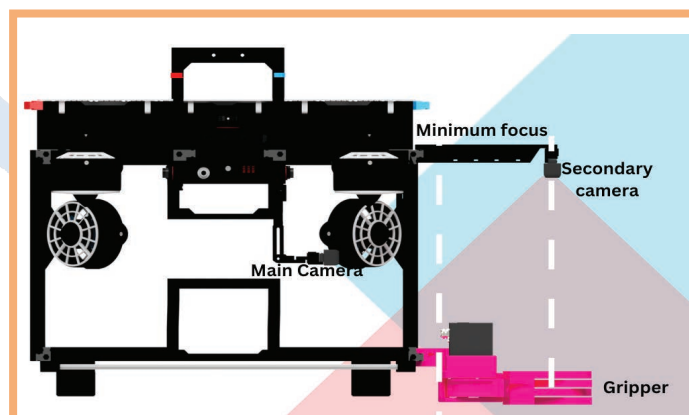


Fig. 7: Systematic Process for Tool Development

Constant referral to the ROV tool layout and mission program helps optimize components and reduce unwanted impacts to the rest of the system. Such reviews also help identify parts that should be standardized and made modular. When necessary, ROV tool layout and mission programs can even be modified if advantageous.

Standard Mounting System 5 (SMS5)

The Standard Mounting System 5 (SMS5) is Fish Logic's fifth generation 3D printed quick mounting system which is used for the attachment of tools and cameras to the Siren ROV.

SMS5 consists of a detachable clamp and a rail.

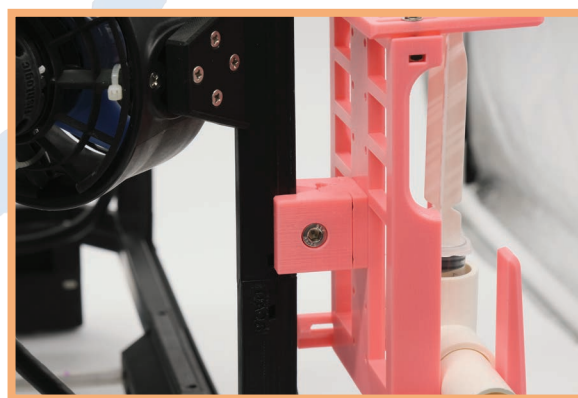


Fig. 8: Syringe Arm Attached to ROV with SMS5

Each SMS5 clamp can be secured onto a SMS rail using only a single M5 screw. All ROV tools are SMS5 compatible by incorporating the standardized centerpiece of the SMS5 clamp. Improving upon the previous SMS4, SMS5 increases durability by adding a center wedge component to the clamp which creates additional friction through one extra contact surface. This provides a more reliable clamping force

that can account for a larger tolerance for SMS rails with dimensional inaccuracies or excessive wear. The rail is the counterpart of the clamp that allows any tools and camera with a SMS5 clamp to be placed in precise locations along the many SMS rails on the

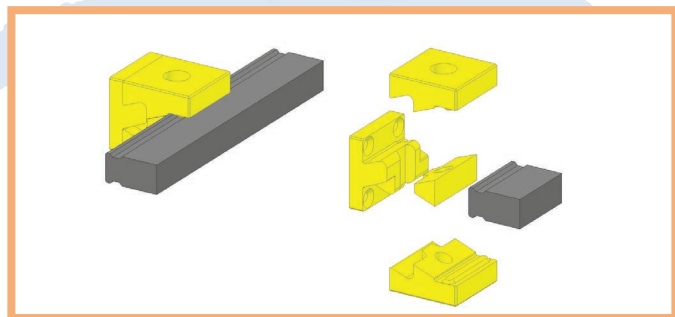


Fig. 9: Standard Mounting System 5

ROV. SMS rails all around the structure of the ROV provide ample mounting spaces for the optimal and flexible placement of tools, as well as reduces the need to change tools due to lack of mounting space. An Aluminium hex rod can be integrated into the SMS rail, allowing the rail to also act as a structural piece of the ROV frame. Since the SMS rail has not been modified since SMS2 from Blazin' Hydra (2018), the SMS5 clamps are backwards compatible with SMS2, 3 and 4. Our standard SMS5 centerpieces are also reverse compatible with SMS4 clamp components. The SMS5 clamp has been tested to be capable of resisting movement of up to 343N in the direction of the length of the rail.

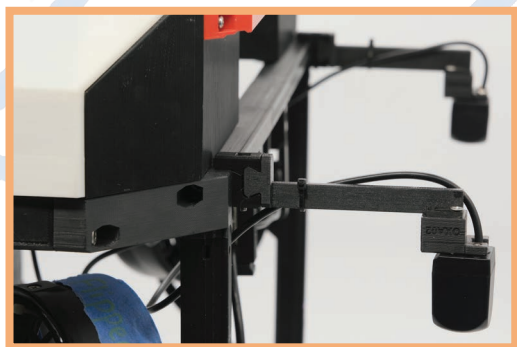


Fig. 10: Camera attached using mSMS5

The Micro SMS5 (mSMS5) clamp is a smaller variant of SMS5 that is used for lighter loads such as cameras and cable clips. The mSMS5 clamp also incorporates the center wedge design that improves durability just like the larger SMS5. This smaller size also allows cameras to be mounted closer to the tools while being lighter and taking up less space on the ROV.

Innovation

Simulator

Due to the lack of local facilities, Fish Logic rarely has access to a pool. This not only bottlenecked the entire development sprint cycle as much of the design and systems could only be tested and validated on the pool access date, but also limited the amount of time the pilot could familiarize themselves with the ROV. In an attempt to solve these problems, Fish Logic created a ROV simulator in Unity. The simulator approximates drag on the ROV by using the triangular mesh of a 3D model of the ROV to calculate the projected area in the direction the ROV is facing. A separate script converts the controller inputs into thrust vectors on each thruster that contributes to the movement of the virtual ROV. Buoyancy is simulated with the calculated volume of the 3D model and known density of ROV parts. This combination provides an approximate physics model with adequate accuracy that can evaluate the performance of the ROV. In addition to getting accustomed to the controls, the simulator has allowed the pilot to test and propose new custom control schemes and tweak ROV speed profiles. The operations team can also run through various scenarios in simulated missions to gain experience in adapting to unexpected situations. By utilizing the simulator, estimated travel time of the ROV between tasks can be recorded. Such data has been taken into consideration when designing the mission program and the tool layout of the ROV to maximize efficiency during the missions. This year, we created a few props to test the pilot's ability to precisely aim when deploying objects and find the shortest routes between task destinations. The simulator also trains the pilot to use finer movements instead of sporadic controls and find the best way to tackle a task.

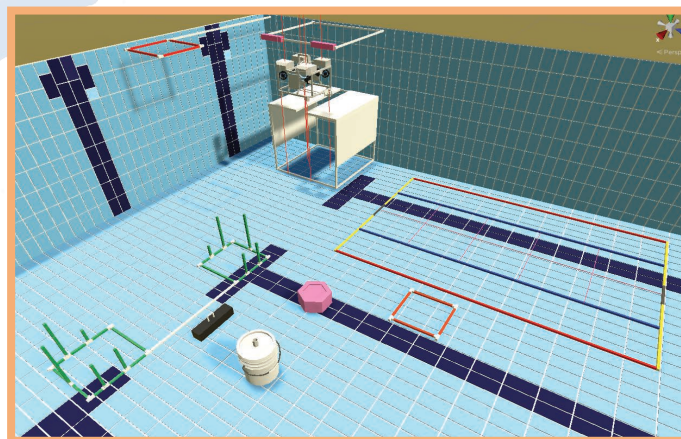


Fig. 11: ROV Simulator with Props in Luso Chinesea Pool

Mechanical

Propulsion and Vehicle Dynamics

Using the Fish Logic ROV simulator, the baseline requirements for the ROV were established by examining the mission tasks, as well as the performance needed to complete all mission tasks within the provided time. Thus determined that the ROV required 5 degrees of freedom (DOF) and a minimum movement speed of 0.6 m/s in all directions.

A Vectored Thrust Configuration was chosen with 4 Blue Robotics T100 thrusters placed at 45 degree steering angle that exert forces in the horizontal plane allowing horizontal translations and rotations around the yaw axis. 2 Blue Robotics T200 are used as vertical thrusters for vertical movement and rotation, with a rated combined lifting force of 56N. The ROV has a recorded maximum speed of 0.63 m/s in all directions.

As the tools mounted on the bottom layer of the ROV changes between every water test, the center of mass of the ROV changes. Since any misalignment between the center of thrust with the center of gravity will produce a pitching motion to the ROV during every lateral acceleration, an adjustable slot was integrated into the ROV structure for the height of the thrusters to be adjusted. This aligns the thrusts of all lateral thrusters with the center of gravity of the ROV, eliminating unintended pitching motion during lateral maneuvering.

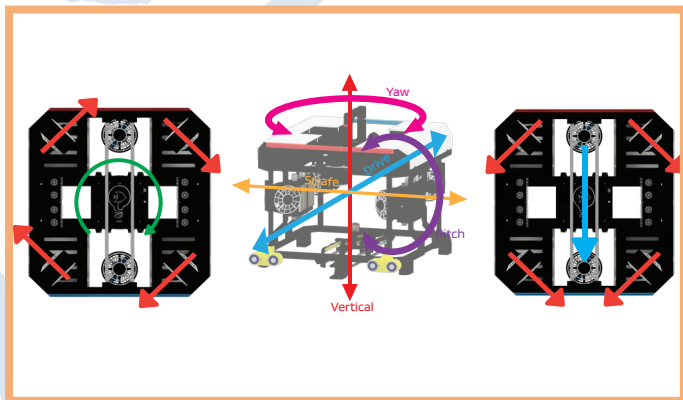


Fig. 12: 5 Degrees of Freedom

Vehicle Structure

The Siren frame is a 3D printed structure made of PLA+, strengthened by 7mm aluminum hex rods. SMS5 rails have been integrated along the edges of the frame to act as a strong structural support and allow for tools and cameras to be freely mounted

anywhere along the rails, granting greater flexibility. The extensive use of 3D printing has enabled the layout and placement of the ROV tools, thrusters and other components to be designed ahead of the

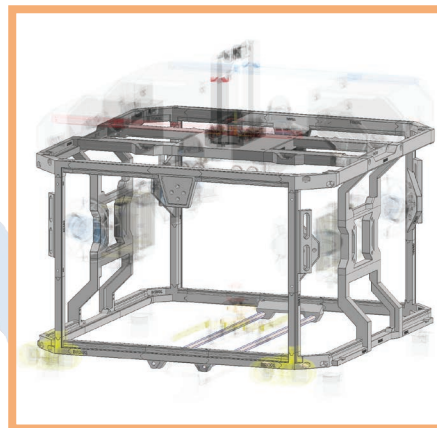


Fig. 13: Siren Frame

ROV structure. Structural parts of the ROV can be specifically designed to provide additional functions such as the aforementioned tool mounting as well as cable management since geometrically complex structures can be manufactured with relatively small increase in manufacturing difficulty thanks to their 3D printed nature. By deciding on the layout and placements before designing the frame, the finalized design of the ROV structure has highly optimized positions for the onboard electronic hubs, cameras, thrusters and floats, freeing up the bottom of the ROV for tools and ballast. Safety principles have been incorporated into the frame's design. For example, a safety factor of 2 has been incorporated when designing structural parts, where all parts of the frame are made to handle twice the amount of load as required during operation. Thrusters and electronics hubs were designed to be placed within the interior of the frame, reducing the chance of external elements coming in contact with potential hazards on the ROV. The adoption of integrating hollow aluminum hex rods within the 3D printed frame has allowed the frame to be much thinner, stronger and reduce hydrodynamic drag for faster acceleration. Individual parts of the Siren frame are made to be modular, allowing parts to be easily upgraded and swapped out in case of damage, therefore improving the ease of maintenance of the ROV.

Material Choice

Materials and manufacturing methods are chosen to complement their use cases and the design of the components to be produced. 3D printing is chosen for its design flexibility. PLA+ from eSUN is the choice of material for all the fused filament fabrication (FFF) 3D printed parts on the Siren. PLA+ is a blend of PLA. It has a higher elongation at break and is therefore less likely to crack, while still having ease of print and print quality offered by PLA. An additional benefit of using PLA+ is that it is a

biodegradable thermoplastic which has less impact on the environment as it can be broken down by bacteria after it has served its purpose.

Materials	PLA+	PLA+	SLA Resin	Aluminum	Steel
Density (g/cm ³)	1.24	1.24	1.18	2.7	7.84
Elongation at break (%)	5	29	6.2	3	15
UTS (MPa)	37	60	91	90	420
Use cases	Aesthetic prints such as the modules on the Control Box due to higher dimensional accuracy and increased color variety	Main material used in most structural parts and tools of the ROV due to its greater strength	Thruster ESC Housing due to complex prints needing higher precision and watertightness	Aluminum hex rods for strengthening ROV structure	Steel laser cut plates are attached to tools that interact with electromagnets

Table 1: Material for ROV

Aluminum rods are also integrated into the frame to increase structural integrity, resulting in a thinner yet stiffer frame. Mask Stereolithography (MSLA) resin 3D printing is used to make small precise parts such as the thruster electronics housing as it is able to print with a small layer height for smooth outer surfaces. MSLA is also used for printing thruster guards as its smooth outer surface provides better hydrodynamics. Steel plates that are laser cut are attached to all tools that interact with electromagnets. (Refer to table 1 for details)

Buoyancy and Ballast

To ensure ease of use for the pilot, our aim for the buoyancy of the ROV is for it to stay upright and to maintain its depth. To achieve this, the overall density needs to be as close to the density of water as possible. High density PU (Polyurethane) foam is used to provide upforce to counteract the weight of the ROV in water by trapping air in its structure. The foam is cut into blocks of a few standard sizes, which can be fitted into any of the six foam chambers situated at the top corners of the ROV to provide a high center of buoyancy.

For an accurate measurement, our team uses the most direct and practical method: weighing the ROV in water using an electronic scale. To offset the ROV weight in water, the volume of float required to displace the amount of water to achieve neutral buoyancy is then calculated with the density of PU foam. The size of the foam chambers is designed to house enough volume of foam to offset the weight of the Siren with additional capacity to account for extra payloads. Fine adjustment of buoyancy can be done swiftly by adding or removing standardized chunks of foam from the foam chamber via a flip-up lid that can be opened single-handedly by pressing the magnetic button latch.

25mm stainless steel ballast balls (64g each, 0.55N

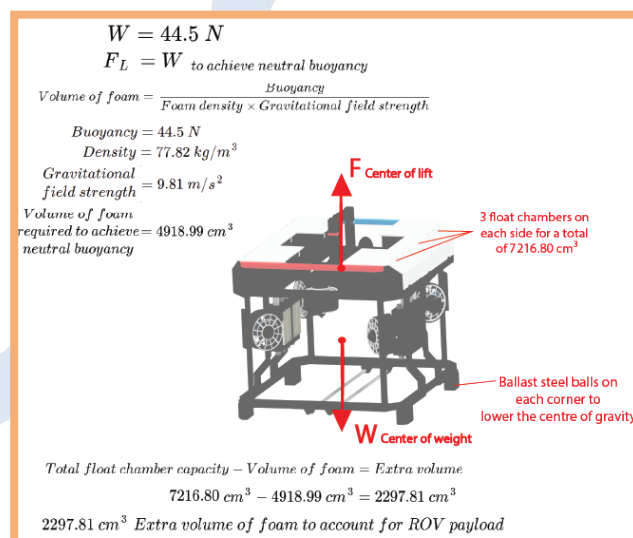


Fig. 14: Weight And Buoyancy Diagram

weight in water) are added to the bottom of the ROV in each corner. This lowers its center of gravity while additionally supporting it when landing on a surface, allowing it to roll on the pool bed. The distance between the low center of gravity and high center of buoyancy produces a restoring force so that the ROV tends to stay upright in and above water. This combined with its neutral buoyancy results in a relatively stable ROV that achieves the above criteria.

Electrical system

SID

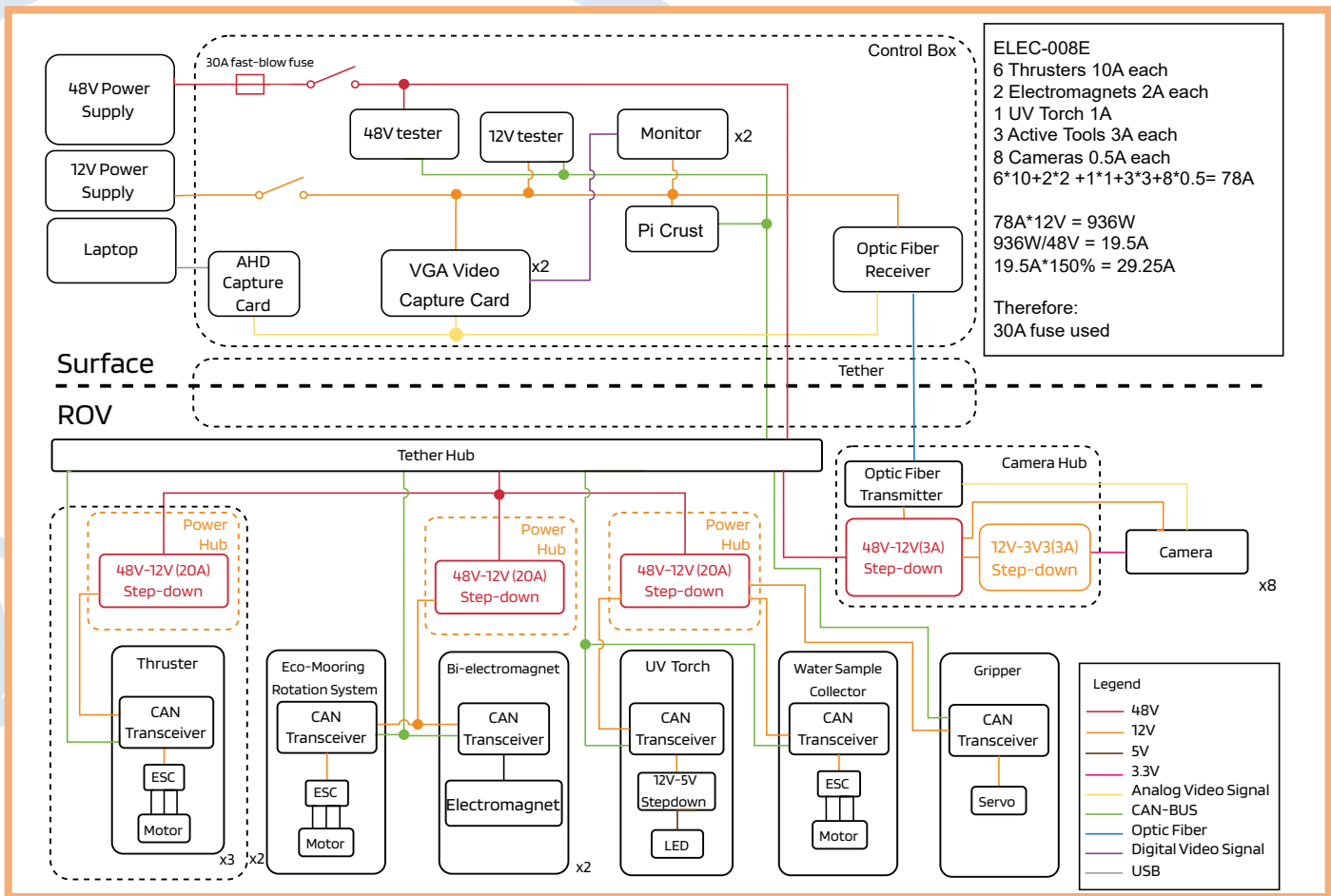


Fig. 15: SID

Power Distribution

The Siren operates at 48V DC. Since the ROV thrusters and tools operate at 12V, 4 Power Hubs are used. Each hub is capable of handling up to 20A in current, converting 48V to 12V to provide the required operational voltage to the devices. This reduces the current load of each Power Hub and enables the power distribution system to be more modular and decentralized as there are less devices connected to each hub. The Camera Hub has an internal 48V to 12V step down, independent from the four 48V step down modules used for thrusters and payload to prevent noise induced from thrusters affecting the analog cameras. Power Hub

Each of the four Power Hubs serves as a 48V to 12V step-down. To handle the maximum current draw of 18A needed to power up to 3 thrusters at the electronically limited max thrust, an off-the-shelf step-down converter rated at 20A was chosen. As each Power Hub is attached to the ROV using 2

screws, a replacement can be swapped quickly in the unfortunate circumstance of a Power Hub failure.

Tether

The power tether uses a pair of 4mm² RVV cables rated for 22A. They are chosen as they contain a PVC sheath layer outside which can enhance the mechanical strength of the wire and protect the cable from corrosion and mechanical damage. The signal tether uses a 0.15mm² TRVVP cable which is a shielded twisted pair cable. It is chosen due to its immunity to interference. The twisting helps with canceling external electromagnetic interference and reduces cross talk between neighboring pairs. The shield also reduces the effect of electrical noise on the signal. An optical fiber is used as part of the tether due to its light weight, flexibility and low susceptibility to noise, making it ideal for transmitting

camera signals. It also reduces the weight of the tether from 0.91kg (20m ethernet cable) to 0.28kg. Our team has considered the use of a single neutral buoyant tether with a continuous foam sheath, however the foam sheath is too rigid as kinks on the tether would cause the ROV to rotate in water. A soft sheathed PVC cable (classified as RVV) is used for

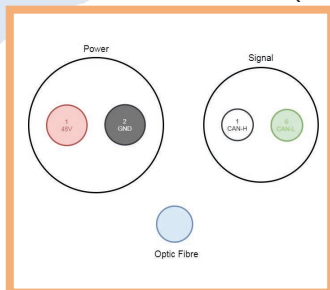


Fig. 16: Tether Cross

power delivery, and a soft shielded PVC (classified as TRVVPS) is used for the signal tether. Stress relief of the tether on the ROV is secured by a tether clamp that constricts the outer insulation of the power cable to prevent any mechanical forces applied to the cable from being transferred to the tether connector, potentially causing electrical terminations within the connector to break. This is reinforced by the use of a metal rope thimble that is constrained tightly to the handle, forming a loop to ensure no forces are transferred to the signal tether and optic fiber cable. The power tether is intentionally made shorter to provide slack for the signal and optic fiber cable.

Tether Management

The rules and protocols for tether management are reviewed by the tether managers in the operations team and abided by during the mission run.

Protocols for tether management

1. Never pull on the tether.
2. Keep the workplace clean and organized for a safe and hazard free environment.
3. Tether is turned in the same direction as the ROV, minimizing the number of tether turns while operating to enhance maneuvering. Remove all kinks on tether before submersible recovery.
4. Be observant of obstacles located near the submersible that have the potential to snag the vehicle or the tether.
5. If the vehicle runs out of tether, there may not be enough slack to allow an easy turn around to follow the tether out. In this case, reverse vehicle direction to generate slack and turn the vehicle around to manage the tether.
6. When operating in an area containing obstructions or obstacles that could snag or foul the tether, the pilot should remember the route taken to get to any one position and the same route should be taken during extraction.
7. If the tether does become entangled, do not pull the tether to free it.

Connectors

The connectors on the Siren are standardized using Weipu connectors as they allow components to be connected to different sockets of the same type. Signal and power are both integrated into a single connector to allow for a quick swap of components. To prevent accidental mismatched connections, connectors of different pins are used for different components. 6-pin Weipu SP13 connectors are reserved for connection to any of the Power Hubs as the manipulators

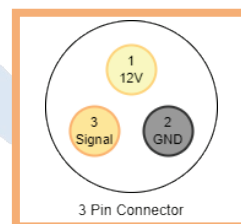


Fig. 17: 3 Pin Connector

and thrusters are all in 12V. 7-pin Weipu SP13 connectors are used solely for 48V components; 4-pin connectors are used for cameras, 3-pin for the brushless motors present in active tools, and 5-pin Weipu SP13 Connectors for PWM components such as the gripper.

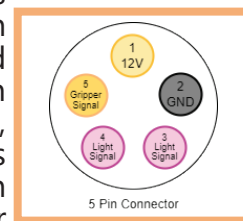


Fig. 18: 5 Pin Connector

Since each pin is only rated 5A of current, 2 pairs of pins are used for power and ground in the 6-pin connectors to allow manipulators and thrusters to get sufficient current.



Fig. 19: Universal Connectors

Although the connectors claim to be waterproof, the back of the connectors are filled with epoxy and silicone grease as a further redundant safety measure to prevent water from leaking into the connector.

Camera System

The Siren is equipped with 8 cameras, 2 of which serve as main cameras while the other 6 act as secondary ones. The main cameras are located at the front and rear of the ROV, one for each end, and are used for primary navigation as the ROV is capable of inverting the driving direction. The secondary cameras are mounted on the SMS rails around the ROV to provide extra viewing angles for mission-specific tasks. Analog High Definition (AHD) cameras



Fig. 20: AHD Camera

are used instead of regular 480p analog cameras and digital cameras due to their ability to achieve 720p resolution while still maintaining a low latency. The increase in resolution provides extra clarity for the pilot and allows farther objects to be easily distinguished during navigation. Since the cameras are IP68-rated waterproof vehicle reverse parking cameras, they only require a preventive layer of UV resin to seal the outer shell seams. Each camera has a 4-pin connector to be plugged into the camera hub to receive signal and power. The Camera Hub, placed at the center of the ROV top layer, digitizes all 8 channels of camera feeds and uses a transmitter to send a multiplex signal through an optic fiber tether to the surface. With the ability to transmit all 8 camera signals simultaneously, it has removed

the need for camera switching onboard the ROV, allowing the pilot and the team to have access to all camera angles at the same time. The use of optic fiber transmission, along with a separate internal step-down module for camera power was chosen over copper wires due to its property of significantly reducing the noise impact from thrusters and the surrounding environment. The optic fiber receivers convert multiplex camera signals received from the ROV back to the 8 individual analog channels. The channels are put through 2 VGA video processors that convert analog to digital video signals. These signals are then sent to the 2 monitors to provide the pilot with 4 different camera views on each monitor.

Control System

CAN Bus System

CAN (Controller Area Network) is the messaging protocol used in the Siren that enables onboard components that are on the same bus network to communicate with one another. The CAN bus system is paramount in achieving standardization on our systems. Any CAN bus device can connect to any socket along the bus network. This enables hot swapping and allows Siren's configuration to be flexible and modular. This contributes to the ease of maintenance and provides a path for future upgrades and expansions.

The CAN bus system was designed with automobiles in mind in order to minimize the amount of electrical wiring used. As a result, the protocol includes error checking and is less vulnerable to noise. These features make it optimal for communication over long distances, and therefore it is perfect for Siren's 20m long tether.

CAN bus was selected over other alternatives mainly for its durability and robustness. Previously with I2C, which was implemented in Fish Logic's Electro Stargazer, the number of errors would increase as messages are sent more frequently, leading to occasional program failures. These stability issues arose significantly less after switching to CAN bus.

The Siren's CAN bus network runs through the tether to the control box, allowing a direct connection to all nodes in the network without needing an onboard translator, which would have created additional complexity and increased potential errors. As no existing implementation of CAN bus communication electronics are small enough to be embedded onto the thrusters, Fish Logic decided to carry out its

own implementation in the form of a BeetleCAN, a custom CAN transceiver designed small enough to fit onto a thruster and placed in each node in the network.

CAN Bus Software Protocol

The primary requirement for the ROV's CAN bus Network is the ability to support communication of up to 20 nodes. The 11-bit identifier of the messaging frame specifies the level of broadcast. The first level is the node level, where the messages are specific for one node. This requires each node to have a unique ID in order to receive the intended messages and ignore other messages by utilizing the mask and filter in the CAN transceiver. The class level allows messaging to all nodes in the same class. Nodes are classified into classes based on their functions (e.g. thrusters, electromagnet and mechanical gripper).

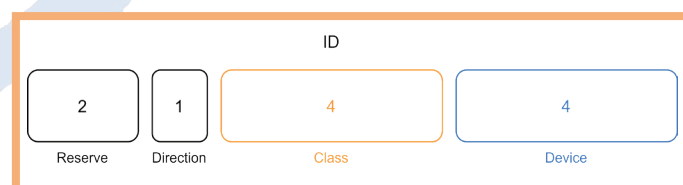


Fig. 21: CAN Bus Messaging Frame

By having a global level that is able to broadcast messages to all nodes, it allows us to perform role calls to ensure all nodes are responsive, as well as make requests for the onboard temperature and voltage readings from all nodes on the network. Table 2 shows a complete chart of which ID's to use to specify levels of broadcast among various classes of devices. The ID is followed by the data, it takes up 1 to 8 bytes of the messaging frame. The first byte is reserved for the command function that the node

should perform. This will be unique to every class. For example, 0x20 indicates the thruster spin command, with the second and third bytes reserved to indicate the speed and direction of the spin.

As messages are being broadcasted along the whole network, bus contention occurs when two or more CAN controllers start initializing messages at the same time, which may cause thruster movements to stutter, or even stop transmitting messages completely. CAN bus resolves this conflict by having every node that wants to transmit monitor the bus, and compare their IDs to see who has the more dominant bit (0) and the less recessive bit (1), to which the dominant transmitter wins the arbitration. This action is performed on the whole CAN bus network until 1 transmitter is left, while other potential transmitters wait until the bus is free. Thrusters are prioritized for more smoothness in control, resulting in the thruster IDs set to have more dominant bits (0). Messages that are sent back from nodes, such as temperature and voltage data, should be less prioritized compared to the messages sent to nodes.

Class	Thruster	Electromagnet	Motor	Light	Servo
Global Level	0xFF	0xFF	0xFF	0xFF	0xFF
Class Level	0x1F	0x2F	0x3F	0x4F	0x5F
Node Level	0x11-0x1E	0x21-0x2E	0x31-0x3E	0x41-0x4E	0x51-0x5E

Table 2: CAN Bus ID Structure

CAN Bus Thruster

Thrusters used on the Siren are similar to regular thrusters. However, an additional microprocessor is added to allow each thruster to process their own information, transceive through the CAN bus network and improve both adaptability and maneuverability.

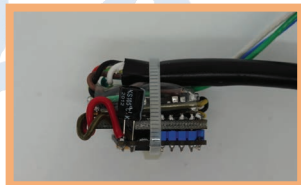


Fig. 22: CAN Electronics and ESC

Control Program

The ROV control program written in Python consists of independent, loosely coupled modules, each serving a specific purpose. Communication between

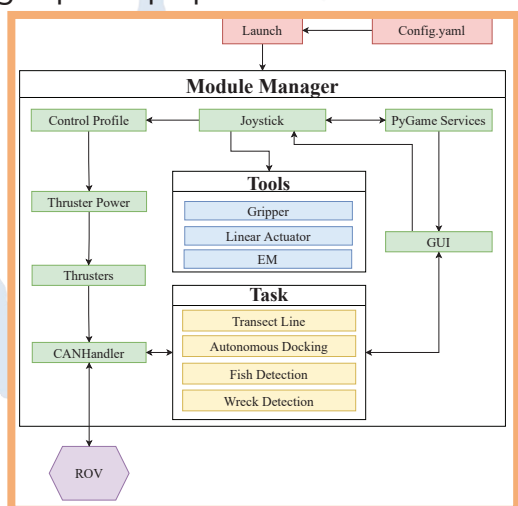


Fig. 23: Module Diagram of Control Program

modules is performed by the PyPubSub library so that modules can broadcast messages while subscribing to specific topics. Python is a synchronous language by nature. However, since many different modules need to run together for the ROV to function, the Python library Threading is used to run modules in parallel.

PyGame, a library which is used for both Joystick and GUI modules, cannot work across multiple threads. The problem was later understood that the PyGame event handler had to be processed in the same thread as the PyGame display module was first initiated. A separate module named PyGameServices was used to instantiate all PyGame modules and process all event handlers in the queue in the same thread. The initiated PyGame modules are then shared between the GUI and Joystick modules.

Thruster Power Mapping

To convert pilot input into thrust values, the control software receives movement command inputs from the joystick module and processes the input values

$$\mathbf{A}_{IN} = \begin{pmatrix} Drive \\ Strafe \\ Yaw \\ Vertical \\ Pitch \\ Roll \end{pmatrix}$$

into a 6 dimensional vector with elements ranging from -1 to 1.

The Moore-Penrose pseudoinverse is used in calculations to map ROV movement direction to the thrust power of each individual thruster.

By measuring the position and direction relative to the ROV's center of mass, a matrix can be constructed that represents the contribution of each thruster to each DoF.

$$B = \begin{pmatrix} D_{FL} & D_{FR} & D_{BL} & D_{BR} & D_{UF} & D_{UB} \\ S_{FL} & S_{FR} & S_{BL} & S_{BR} & S_{UF} & S_{UB} \\ Y_{FL} & Y_{FR} & Y_{BL} & Y_{BR} & Y_{UF} & Y_{UB} \\ V_{FL} & V_{FR} & V_{BL} & V_{BR} & V_{UF} & V_{UB} \\ P_{FL} & P_{FR} & P_{BL} & P_{BR} & P_{UF} & P_{UB} \\ R_{FL} & R_{FR} & R_{BL} & R_{BR} & R_{UF} & R_{UB} \end{pmatrix}$$

Key
D – Drive *FL* – Thruster Front Left
S – Strafe *FR* – Thruster Front Right
Y – Yaw *BL* – Thruster Back Left
V – Vertical *BR* – Thruster Back Right
P – Pitch *UF* – Thruster Up Front
R – Roll *UB* – Thruster Up Back

The Combined thrust to reach the desired effect of B can be calculated as follows:

$$\begin{pmatrix} A_{IN} = BX_{OUT} \\ B^{\dagger}BX_{OUT} = B^{\dagger}A_{IN} \\ X_{OUT} = B^{\dagger}A_{IN} \end{pmatrix}$$

Where:

$B^{\dagger}$ is the Moore – Penrose pseudoinverse of matrix *B*
 X_{IN} is the 6x6 matrix mapping thruster contribution and DoF
 A is a 6 dimensional vector containing the movement command
 X_{OUT} is the resultant 6 dimensional vector containing required power of each thruster

$$X_{OUT} = \begin{pmatrix} FL \\ FR \\ BL \\ BR \\ UF \\ UB \end{pmatrix}$$

Calculation with matrices and vectors is chosen as opposed to hard coding the calculations for each thruster's power individually. It allows the code to be more concise and can easily adapt to changes in the thruster configuration by editing the preset matrix. This algorithm can factor in the imbalance of force or asymmetrical placement of thrusters to provide more accurate movements thus any command from the pilot will result in the desired movement of the ROV.

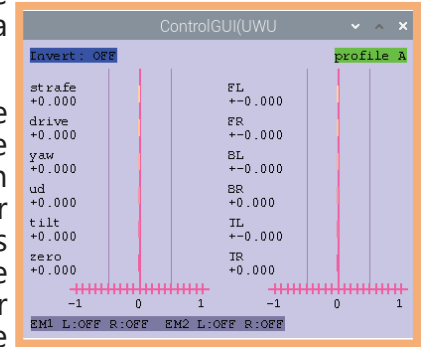
GUI

The Graphical User Interface (GUI) is designed to aid the pilot in the operation of the ROV, and help the team with troubleshooting the control program and ROV systems.

The GUI elements are designed and placed

strategically to be viewable at a glance. Clear indicators are displayed for each control of the ROV, whether it be done with printed text or a graphical indicator.

Status bars are used to display the thrust values of each thruster. Gripper movement prompts that indicate the static, opening or closing states are displayed to help diagnose issues with the input interface or Control Program.



The GUI utilizes the PyGame Library in order to draw the graphical elements. PyGame is chosen for its ease of use and highly modular architecture, with only GUI and Joystick modules utilized to minimize performance impact. The GUI communicates with other modules with PyPubSub and subscribes to all the data that needs to be displayed.

Control Box

The Siren Control System contains two 22-inch 1080p monitors, allowing the pilot to view all 8



Fig. 25: Externals Of Control Box



Fig. 26: Internals Of Control Box

cameras simultaneously. An optic fiber receiver, 2 VGA video processors, an AHD Capture Card and a Pi Crust housing a Raspberry Pi 4 can be found under the monitors. The Pi Crust runs the Control Program and transmits CAN bus signal to the ROV via the

Power and Signal port on the control panel, allowing commands such as maneuvers to be executed, while displaying the status of the ROV using a custom GUI. The AHD Capture card is used to output camera footage to a laptop via an USB cable for performing Photogrammetry and 3D-model generation. A tester for 12V and 48V electronics is also available on the control panel to conduct tests on electrical parts quickly and easily. With power routed around the Control Box, a built-in kill switch allows the power of the ROV to be completely cut off in an emergency. No AC power was used in the Control Box.



Fig. 27: Xbox Controller

Mission Specifics

Gripper V3

The Gripper V3 is Fish Logic's third generation double-pivot claw. It is designed as a general purpose tool with a multipurpose grip surface for the transportation of the tent, removal of marine growth, placement of camera, and connection of the power connector. As the most versatile tool, the gripper could also be used to support other tasks with the discretion of the pilot.

The new gripper uses the ROVMaker waterproofed servo motor instead of the brushless motor with



Fig. 28: Gripper V3 (closed)



Fig. 29: Gripper V3 (open)

The grip faces and jaws are designed to be modular which allow for fast repair and hot swapping. The grip faces could grasp objects up to 160mm apart, and since it's overlappable, even small and thin objects

dace fry, and anchoring the solar panel. The choice to

could be grabbed on. The grip faces are also casted with a 3mm layer of silicone rubber to increase friction and have a tighter grip on any object. Silicone rubber was chosen over foams used on previous designs as it is not significantly affected by water pressure.

The previous gripper design (V2) encountered several problems especially lock ups due to friction. The friction is caused by the threaded block and the threaded collar digging into the edges of the grippers, which outmatched the force provided by the brushless motor. As a result of these lockups, the edge plates experience a lot of force, causing them to break easily. Other than the problem of lock ups, the previous gripper was also difficult to maintain due to its complexity in assembly. Furthermore, improvements of parts couldn't be made without first changing the thin metal plates.

The opening speed and closing speed of the Gripper V3 is 46.7°/s. The jaws could open up to 70° and thus give a maximum distance of 180 mm between each jaw.

Bi-Electromagnet

The Bi-Electromagnet (BEM) is a general purpose tool used in deploying the lift bag, acclimating the



Fig. 30: Bi-Electromagnet

use electromagnets was due to its simple working principle with no moving parts. This makes it easy to manufacture, maintain and use, while being reliable and versatile for many tasks. Each BEM consists of two individual electromagnets which are positioned next to each other. Siren, equipped with two BEMs, has a total of four electromagnets that can be individually turned on and off, thus giving it the capacity to hold four objects simultaneously. The center point between the two electromagnets are spaced apart to the width of the U-bolt present on the patch for the fish pen in order to ensure a secure contact. The two electromagnets on the BEM are rated for a combined force of attraction of 36kg. LEDs are embedded into the BEM to indicate the status of the individual EMs.

An onboard H-bridge motor driver allows polarity of the electromagnet to be reversed to counter magnetic hysteresis, where light ferrous objects that are magnetized remain attracted to the BEM even when the electromagnet is switched off. Multiple mounting points are located around the BEM, which allows tools such as the BEM clasp to be mounted on it.

Task 1

Carabiner Latch

Attached to the Bi-Electromagnet, each Carabiner Clasp is responsible for holding and anchoring a mooring connector of the solar panel array to the designated U-Bolts swiftly and reliably. During operation, the mooring connectors are encapsulated within the tool, eliminating all twisting motions

and allowing the pilot to precisely aim the anchors during installation. The connectors are locked within the tools using a flap, which is attracted to the activated BEM, and studs which extend into the PVC coupling of the connectors. A funnel is also attached under the tool to prevent sliding motions when attempting the task, which would cause the carabiner to slip off of the U-Bolt.

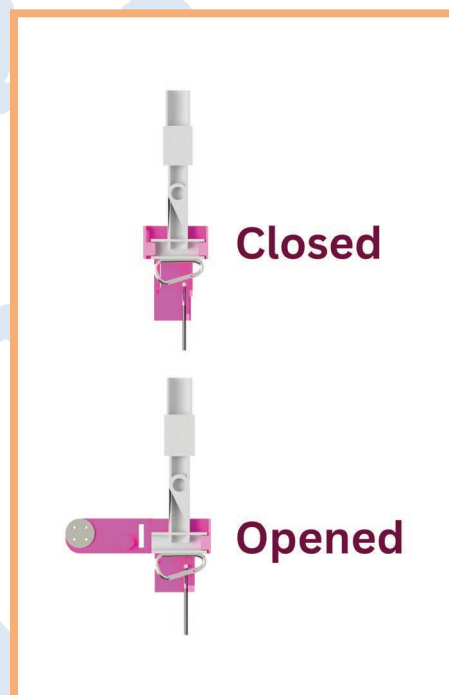


Fig. 31: Carabiner Clasp

A slot on the funnel allows it to extend and retract, preventing the funnel from hitting the ground whilst also allowing for a larger contact surface with the anchor point.

When attempting this task, the pilot would guide the U-Bolt into the funnel and under the carabiner. Once the U-bolt is aligned with the bottom facing connector, the ROV moves downwards, pushing the

carabiner against the U-bolt in a constrained manner, allowing for only a single direction of movement. The gate of the connector is forced open and the U-Bolt is slipped into the carabiner. Once the connector is anchored, the electromagnet is turned off and the flap is opened, allowing the ROV to move on to the next mooring connector.

Power Connector Cap One Way Latch

A simple magnetic one way latch design consisting of two attracting magnets was chosen to retrieve the power port cover. A long M3 screw is used as the latch's shaft and two hooking points that slant inwards help guide the power port cover into the latch. To make it easier to hook and trap the power port cover, the latch was intentionally elongated. The tool would first go under the power port cover through the 40cm rope before slowly rising up, guiding the power port cover into the latch using the hooking points. The weight of the cover pushes the one way latch open, securing the power port cover inside the one-way latch.



Fig. 32: Power Connector Cap One Way Latch

Task 2

Water Sample Collector

For the ROV to successfully retrieve an eDNA sample of the coral head, the ROV is equipped with a sample collector that draws water above the coral head; this collector uses a 50mm syringe to store the water sample and is mounted on the side rail of the ROV for easy piloting.

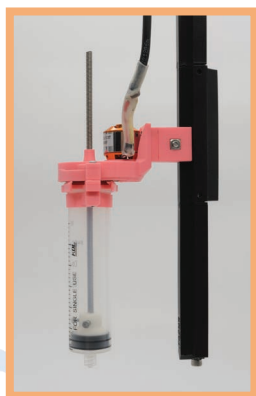


Fig. 33: Water Sample Collector

To collect the water sample, the sample collector uses an M5 threaded rod connected to a 0.5 module 84 tooth gear to create linear motion. This gear is mated with a 12 tooth driver gear driven by a brushless motor. This 1:7 ratio set up is to create enough torque to pull and push the plunger. This rod is attached to a 3D printed adapter which is connected to the rubber plunger of the syringe so that when linear motion is created, the plunger is moved up or down along the M5 threaded rod.

A brushless motor is chosen due to its reliability and modularity, allowing the water sample collector to be easily adapted to our present system.

UV Torch

The UV Torch is a semi-active tool used to irradiate diseased areas of coral head. It simulates UV light with a LED module controlled via a CanBeetle module; power is supplied by a 12V to 5V mini-360 buck converter. The UV Torch is integrated with a SMS5 Clamp, allowing it to be mounted at the optimal position on the ROV. This allows the pilot to aim the torch at the diseased area with ease.



Fig. 34: UV Torch

Syringe Arm

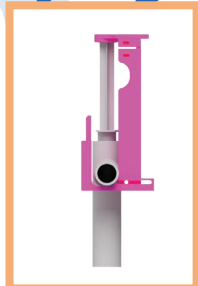


Fig. 35: Syringe Arm

The Syringe Arm is responsible for transporting a syringe with probiotic fluid and injecting it into the tent. This tool consists of two arms and a roof. The arms hold onto either side of the syringe and prevent it from falling, whereas the roof is designed with an indentation that allows the top of the plunger to fit exactly in it, suppressing any chance of the syringe disengaging during injection.

When injecting, the tool is pressed down, allowing the roof to press into the plunger. Both the arms and the roof are adjustable vertically and horizontally to allow a syringe of any shape and size to fit within.

Eco-mooring rotation system

The Eco-mooring system consists of a Bi-Electromagnet (BEM) rotation latch attachment, a rope holder and a float cap. The BEM rotation latch is attached to the BEM via a screw for the placement and installation of the eco-mooring. A flap is connected to a hinge in a way which allows it to swivel freely; the metal plate on top of the flap enables it to act as a latch that can be activated and deactivated using the BEM. The eco-mooring is installed by a spinning reel mechanism attached to the latch, driven by a brushless motor through a 1:4 ratio gear train. Foam brackets are placed on both the core and the flap to provide a firm grip on the eco-mooring. After installation, the BEM will be turned off, releasing the latch and allowing the flap to swing open, thus enabling the eco-mooring to be deployed.

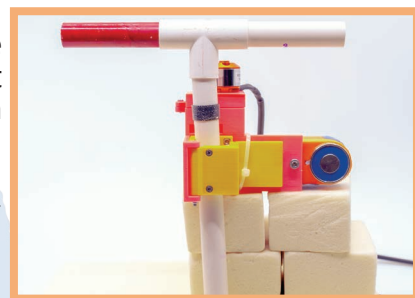


Fig. 36: Eco-mooring System

While the eco-mooring system can be held with just the BEM latch, its ropes would float around, blocking the camera's view and tangling with the other tools on the ROV during deployment. The rope holder is a simple U-shaped bracket padded with foam and keeps the eco-mooring in place while preventing its ropes from moving during transportation. The Float Cap is an inverted cup that takes advantage of the buoyant property of the eco-mooring's float, keeping it in place until the ROV deploys the eco-mooring.

Dace Fry Deployer

The Dace Fry Deployer is a rectangular cage used to acclimate and release Northern Redbelly Dace fry in the safe release habitats. The Deployer is activated



Fig. 37: Dace Fry Deployer (open)



Fig. 38: Dace Fry Deployer (closed)

by a pivoting trapdoor attached to a trigger with a metal plate on it. This allows it to be picked up by the ROV using the BEM. By pulling the trigger, the trapdoor rotates 90 degrees, opening the trap and releasing the fry. Two 50g brass weights are mounted on to the base of the cage, preventing the cage from toppling over during actuation. To prevent the accidental release of the fry during transportation, semi circular metal plates are attached to the trapdoor, allowing it to be securely held in place by the ROV using the BEM.

Lift Bag

The Lift Bag has a diameter of 35 cm, giving it a maximum capacity of over 22L of air (220N of upthrust) and the ability to transport the 12kg container from the seafloor to the shore. The Lift Bag uses a one-way latch to hook on to the container.

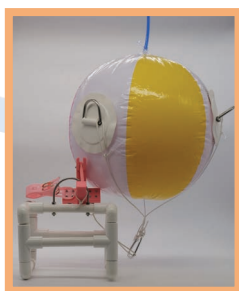


Fig. 39: Lift Bag

Once the hook latches on to the U-bolt of the container, repelling magnets on the one-way latch keeps the gate closed and prevents the Lift Bag from disengaging from the container. The two ferrous metal plates attach to the Bi-Electromagnet on the ROV to allow for ease of transport. When the Lift Bag is released from the ROV, air is pumped into the Lift Bag, allowing it to float up to the surface and be pushed to shore for collection.

Ponyo

Ponyo is a vertical profiling float designed to complete two vertical profiles. The buoyancy engine

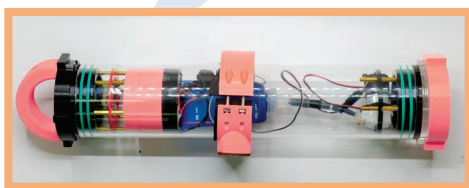


Fig. 40: Ponyo Vertical Profiling Float

uses a gear pump to transport water between the pool and an internal bladder. 3D-printed frames help support and secure the flexible bladders to prevent the tubes connecting to the peristalsis pump from restricting water flow. The gear pump is controlled by a motor driver built into the Romeo BLE mini V2 Arduino. All electrical components are powered by 16 AA alkaline batteries as its power source as it is able to be contained in the tight space the canister provides. The batteries are divided into 2 battery boxes, each with 8 cells connected in series to provide a total voltage of 12V. The battery boxes are connected in parallel, yielding a 4 ampere-hour capacity.

When the float reaches the surface, a float-attached magnet descends around the canister to activate a hall effect sensor within the housing to indicate that the surface has been reached. To detect

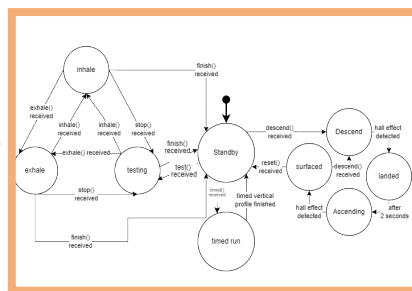


Fig. 41: Ponyo FSM

when the float has reached the bottom, a magnet on a guided rod would hit the bottom of the pool and move up to activate the bottom hall effect sensor.

The Arduino is used in conjunction with an antenna, which allows Ponyo to be capable of communicating with the mission center via radio waves. The inclusion of wireless communication with the Ponyo allows it to be reprogrammed, controlled manually, and relay information to the mission center as soon as it resurfaces.

In-house built Vs. Commercial Components

Fish Logic fabricates in-house parts to achieve a level of innovation and unique functionality that could not be commercially purchased. The components are evaluated with numerous aspects in mind, including its performance, reliability and feasibility for the members to manufacture. Devices with complicated functionality, which exceeds Fish Logic's fabrication capability, such as cameras, monitors and the gamepad are purchased.

With the many new components that Fish Logic aims to improve on, the team first researches for commercially available parts. Parts that fulfill the required specification are compared to the resources and skills required to fabricate it in-house. The features, performance, cost and especially reliability of the parts are extensively evaluated before purchase. When no suitable parts are available, Fish Logic then considers whether the team has the time, resources and the skills needed to fabricate them.

For the case of the thrusters, the commercial T200 thrusters have excellent performance, however, there are no commercially available CAN bus transceiver modules that are small enough to fit within the ESC housing of the thruster, therefore a custom PCB was designed and sent to a manufacturer to be printed. This results in a CAN bus transceiver that is compatible with an arduino microcontroller and small enough to fit within the ESC housing to be waterproofed.

New or Reused Components

Fish Logic strives for continual innovation and brings improvement to all aspects of the ROV. The decision on whether the parts are reused depends on the new features, improvement in performance, effort needed to fabricate and financial cost of the components.

This year's Siren is a minor evolution from last year's Otodus. Since there were only minor improvements done to the basic platform of the Siren over the Otodus, common core electronics were reused. This maintains the commonality between platforms and allows the Otodus to be used as our stable testing platform. As last year's ROV control laptop had CAN bus reliability issues, the new control system reverted back to using the Raspberry Pi 4, thus allowing the Pi Crust from the 2020 Hydron ROV's control system to be incorporated as a reused component.

Mission program

Fish Logic uses the mission program to define the layout of the ROV. The complete mission program is the final verification that all mission tasks can be completed within the time limit of the product demonstration. The order of the mission program is set based on the difficulty and tool combination to have a streamlined sequence of tasks for the pilot to execute. The tool combination between each return to the surface is grouped as a layout that our operation team can reliably set up. Systems engineering is used to make sure all tools are well integrated together without compromising the functions of each other. The final mission program requires 2 resurfaces, thus 3 layouts where all tasks can be completed in the 15-minute time limit.

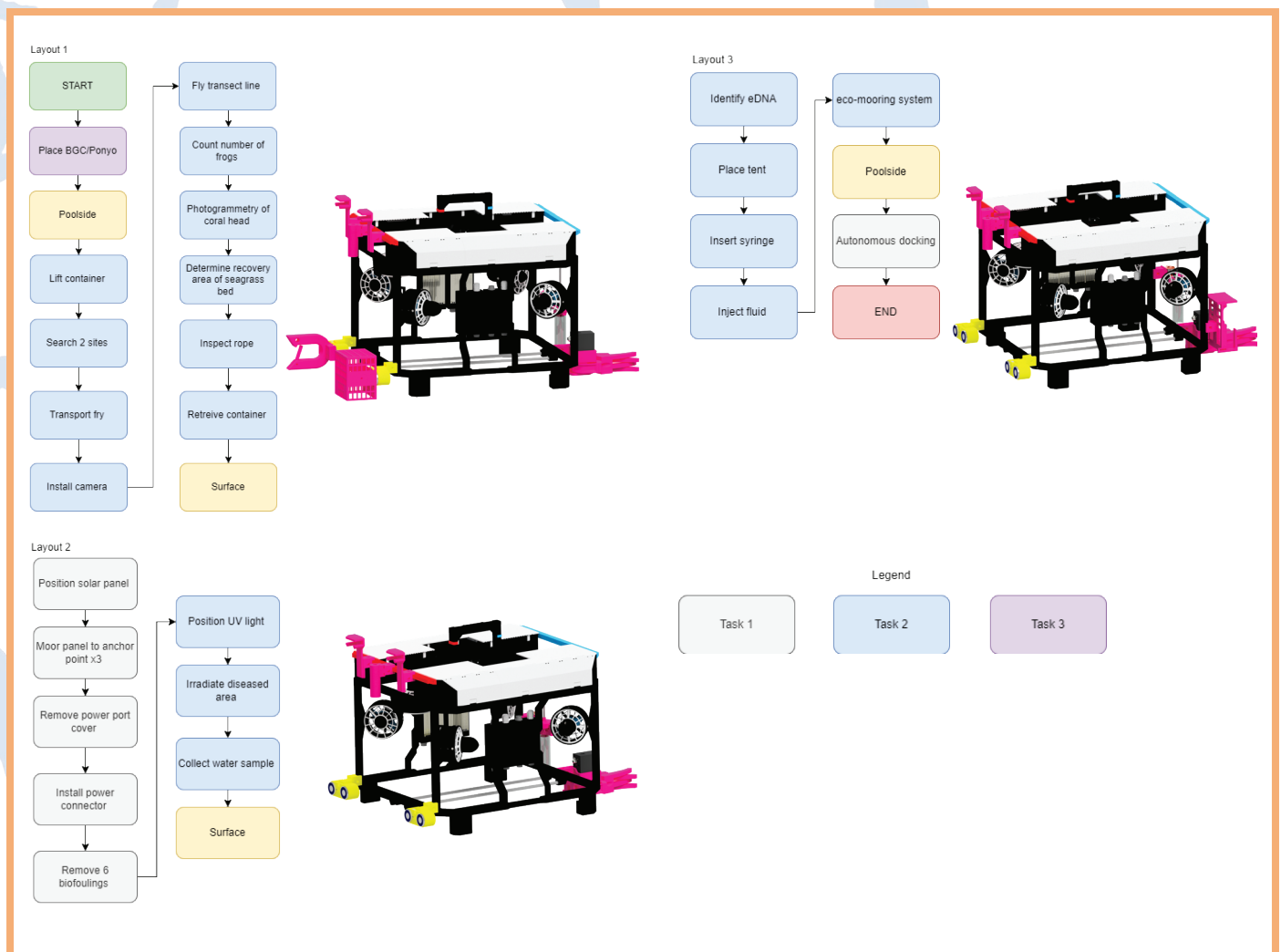


Fig. 42: Mission Program

Testing and Troubleshooting

Vehicle Testing

In regards to testing, Fish Logic testing principles are followed: design a testable system, test like you fly and test what you fly. Rapid design-build-test cycles that maintain a continuous design heritage are to inform the next design by the experience gained.

Documentation and testing processes become more formal as the system matures. Tests are categorized into development, qualification and acceptance tests. Development tests are used to determine hardware capability in excess of requirements and to find weaknesses, such as the 20 meter water pressure test, ultimate strength tests etc. This is to test reliability in different scenarios and gather important data, including: video footage, forces and power draw etc. Qualification tests demonstrate hardware performance limits, where components are tested within the worst case conditions plus the required safety factor. This helps to understand the characteristics of the tools when functioning near the designed limits. Acceptance Tests verify workmanship and functionality. All components are acceptance tested before being certified for use on the ROV. Hardware in the Loop (HITL) verifies hardware-software integration. For every hardware-software change, a full system test is performed as the acceptance test.

Prototyping and Testing

The development of the Ponyo was the best example of evolutionary and integration prototyping used by Fish Logic. Due to the weak structure and strong vacuum created in the previous SeaLift, the syringe system was shelved and a new approach was sought. Originally, a system using a gear pump, an internal bladder and an external bladder was favored and chosen. The technology and conceptual prototype of the Ponyo vertical profiling float was first built using readily and commercially available materials to quickly test the concept of the mechanism and better understand the system. The design used a juice packet housed inside a large 90mm diameter canister to hold 200ml of water. Another juice packet located outside the canister acts as the external bladder. Prototype electronics consisting of AA battery holders, a DC gear pump and a Romeo BLE Mini microcontroller were quickly put together. The water within the internal bladder is pumped out into the external bladder by the gear pump. Each sub-system goes through a Unit Test before assembly, such as testing the bladders and pump for water leakage with a handheld vacuum pump. The electrical system is tested with a multimeter before and after

everything is soldered ensuring the connections and power distribution is as planned. Water seal tests are conducted to make sure the canister and connections are waterproof up to 5 meters. After the qualification of every sub-system, the whole vertical float is fully assembled and integration tested to ensure all units are nominal and functional. It is then tested in an olympic sized (2m) swimming pool to verify functionality. Finally, the conceptual prototype is brought to a diving pool (5m) to be field tested. However, due to the omission of testing and data regarding the gear pump's ability to push water at high pressures, we found out that our Ponyo was unable to resurface after reaching the bottom of the diving pool the hard way.

Thus, a new system pumping water between the pool and an internal bladder with a peristalsis pump was conceived. We made sure to verify the specifications of the peristalsis pump (1.5bar) and ensure that it was able to pump water at great depths using pressure chambers and gauges. Thankfully, a majority of the parts could be reused and the change was minimal. However, we ran through our testing process once again to ensure the capabilities of Ponyo meet expectations.

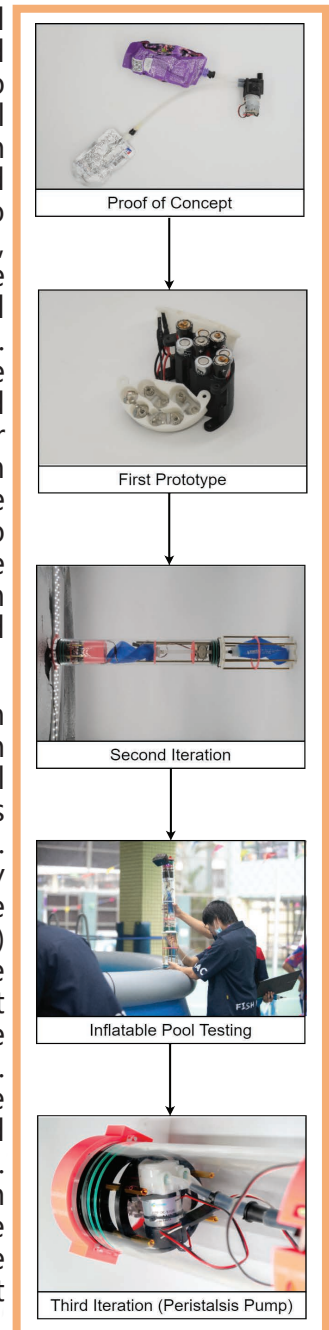


Fig. 43: Ponyo Design Evolution

Through this experience, our team has been reminded that it is critical to not only test and gather data, but also ensure we gather the correct data, lest similar events recurring. We were lucky that no major changes to our system were required, but we must be careful in the future with how and what we test at all levels and the evaluation of our testing processes.

Troubleshooting Strategy

The Gripper V3 was developed as a result of the complexity of the previous design. The previous gripper, Gripper V2, uses a power screw mechanism with a 12-teeth gear connected to a BLDC motor which drives a 36-teeth gear, giving a gear ratio of 1:3. This method of articulation allows for a very strong grip, but is overly complicated and hard to maintain. In addition, this design experiences frequent lock-ups when the center block rams into the edges at high speed while closing or opening to the max. This causes the block to dig into the plastic and tighten the central screw, preventing any further turning motion.

At first, for the 1st iteration of Gripper V3, it was decided that the use of a worm gear to drive two other gears in opposite directions would solve the issues of lockups and complexity. However, this setup created an unexpected problem where the two gears would crash into the casing due to the direction of spin of the worm gear. This force created enough friction to completely lock-up the claws from moving, betraying the initial intention of the worm gear.

Through meetings and deliberation with the team, we settled down on using a servo motor for articulation instead for its reliability and precision. The second iteration was made but problems soon followed as the servo was found to be too tall and would block the camera's line of sight from the claws. The claw design is also too big, significantly increasing the weight of the tool.

After several tests, we found out that by aligning the servo to one side, there would be enough space for the main camera to capture the position of the claws. Though this may cause weight imbalance of the tool, it was not very significant to the degree that it would cause safety or maintenance issues. For the new design of the claw, we took inspiration from the mechanics of universal clamps, where it could grip

onto objects of any shape and size up to a certain length. The claw is made up of fingers which could overlap to hold onto small and thin objects.

The fingers' lightweight and slim design decreases the weight of the tool, but due to the small contact surface area when grabbing an object, it tends to let them slide and fall. At first we tried using foams from previous designs to solve this issue, but we soon discovered that it would not be effective under great water pressure as it is easily compressed. We then tried using crystal tape as the contact surface



Fig. 44: Gripper Finger

due to its sticky surface that allows good grip onto smaller objects. However, for bigger and bulkier objects, it lacks elasticity to provide a large contact surface area.

The last option we decided to use is Ecoflex 00-30 silicone rubber. The silicone rubbers are affected less by water pressure and have a great elasticity that could provide a large contact surface area on objects. This material's high durability under water along with a tensile strength of 200 psi allow it to maintain in perfect shape even in prolonged and extensive use underwater. Scientific tests are performed on pieces of silicone rubber to make sure it suits the conditions and tasks it would encounter.

Through this experience, our team learned that even minute details may have a great effect on the functionality and ease of use of tools. Our team realized that a proper troubleshooting strategy based on scientific concepts and proper testing is always the ideal way of problem solving as this process would have saved time from trying improper inconclusive experiments.

Safety

Safety Principles

At Fish Logic, safety is our number one priority. The safety of personnel, wildlife, and the environment is considered for all operations and decisions carried out during the development of Siren. To achieve this, Fish Logic has adopted the 4 phases of Emergency Management: Prevention, Protection, Response, and Recovery. Prevention measures are taken to avoid potential hazards and accidents. Protection equipment is used when certain exposure to hazards cannot be avoided and can be controlled, such as wearing safety glasses and protective gloves when using a power drill or a sharp tool. Responses should be quick and effective should an unfortunate incident happen. Recovery of the work area back to safety standards after the incident will be a full analysis of how to prevent such problems. New team members are briefed on the safety principles and taught safety procedures required for both in the workshop and at the poolside. Vehicle safety features are considered right at the design stage to ensure the ROV meets all safety requirements.

Vehicle Safety Features

To ensure the safety of the vehicle, the team personnel working around it, and the surrounding environment, several safety measures are taken as shown in the following table.

Safety Features	Description
Light	Lights are installed on the ROV for it to be more noticeable in its surroundings to avoid accidents such as bumping into other ROVs, sea creatures, or even divers.
Thruster Shrouds	IP20 standard compliant 3D printed thruster shrouds are installed onto the thrusters, which prevent fingers from getting near the propellers and sustaining injury.
No Sharp Edges	All exposed edges of the ROV and the modules are chamfered to prevent cuts and injuries during handling.
Handle	A handle is installed on top of the ROV which allows safe and unobstructed handling.
Fast Blown Fuze	Fast blown fuses are used to cut the circuit when the current exceeds the designed limit, which can prevent accidents such as shock due to unintended conduction.
Kill Switch	A kill switch is installed in the control box to cut all power to the ROV in case of an emergency.
Notification LEDs	There are notification LEDs installed on different parts of the ROV to show that the ROV or the electronics modules are powered.

Table 3: Safety Features of ROV

Finance

To balance our finances, Fish Logic requested our mentor to apply for funding based on our planned budget from our school Macau Anglican College (MAC). MAC has provided Fish Logic with generous funding for our current project, the Siren, as well as our previous endeavors. On top of that, Fish Logic has fundraised during our school's open day event by selling our own merchandise, which contributes as our second source of funding.

Budget Planning

As the project commences at the end of the previous competition, a budget is prepared by the team with estimated expenses based on the expenses of the last ROV. The income is estimated approximately to the funding and is included in the budget plan. Once the income has been received, the team verifies that the amount meets our income to see if adjustments to the budget need to be made. The team must stick to the budgeted expenditures by reviewing the budget at the end of each sprint cycle. Additionally, purchasing costs are allocated for the development of tools based on the budget for the next sprint cycle. Whenever receipts are collected, it is entered into the budget. A project costing sheet is compared against the budget to make sure that the capital is used properly and no extra capital is used. The team was able to reduce travel costs as all pieces of equipment are hand carried on the plane. For shipping costs to the local pool, Fish Logic evaluated different trucking services and chose them based on the criteria of cost, service, reputation, and punctuality. Once the truck service was selected, the shipping budget was allocated based on shipping costs and the number of pool access dates.

To incorporate finance planning in Scrum project management, Fish Logic created a budget estimation based on the tasks of each sprint and the release planning at the beginning of the year. At the beginning of each sprint, the CFO allocates the budget set for the current sprint for the team. All spending is tracked in an Excel spreadsheet. At the end of each sprint, the team re-analyzes the efficiency of budget spending, and leftover budgets would be reallocated to other sprints. With Fish Logic's focus on maximizing in-house built parts, the cost of production could be controlled. An example would be the Gripper. During the development of the gripper, the production cost was also considered aside from functionality and weight issues. Due to this, Components with complex mechanisms and needs to be highly reliable such

as the servo and thrusters are purchased. Other parts of the gripper were 3D printed to save on cost. 3D printing also allows us to accurately predict production costs as material consumption and printer running costs can easily be calculated before prints are made. By incorporating finance planning with Scrum project management, Fish Logic was ultimately able to stay within the set budget while simultaneously being able to exploit the benefits of using Scrum project management.

School Name:	Macau Anglican College	Reporting Period:	From:	01/09/2022
Instructor:	Andy Tsui		To:	23/05/2022
Team Name:	Fish Logic			
Income				
Macau Anglican College Grant			50,000.00 HKD	
Funds from selling merchandise			2,200.00 HKD	
Category	Type:	Descrption/Examples	Projected Cost	Budgeted Value
Hardware	Re-used	Blue Robotics Thruster	6,294.54 HKD	0.00 HKD
Hardware	Re-used	Camera	712.32 HKD	0.00 HKD
Hardware	Re-used	Re-used Hardware	17,417.19 HKD	0.00 HKD
Hardware	Re-used	Re-used Electronics	4,398.12 HKD	0.00 HKD
Hardware	Purchased	3D-Printer Filament and Resin	1,324.38 HKD	1,500.00 HKD
Hardware	Purchased	Waterproof Plugs and Enclosure	476.40 HKD	500.00 HKD
Hardware	Purchased	Electronics	1,279.88 HKD	1,500.00 HKD
Hardware	Purchased	Hardware	269.62 HKD	1,000.00 HKD
Hardware	Purchased	To swimming pool	1,457.10 HKD	1,500.00 HKD
Travel	Purchased	Accomodation in Hong Kong	8,000.00 HKD	10,000.00 HKD
Travel	Purchased	Transportation to Hong Kong	2,600.00 HKD	3,000.00 HKD
Total Income:			52,200.00 HKD	
Total Expenses:			44,229.55 HKD	
Total Expenses - Re-used/Donations:			15,407.38 HKD	
Remaining Budget:			36,792.62 HKD	
Total Fundraising Needed:			0.00 HKD	

Table 4: Budget Planning table (All currency in HKD)

School Name:		Macau Anglican College			Reporting Period:		From:	01/09/2022
Instructor:		Andy Tsui					To:	23/05/2022
Team Name:		Fish Logic						
Date:	Type:	Category:	Description:	Unit Price:	Quantity:	Expenses:	Running Balance:	
01/09/2022	Cash Donated	Funds	Funds from Macau Anglican College	50,000.00 HKD	-	50,000.00 HKD	50,000.00 HKD	
25/03/2023	Cash Donated	Funds	Funds from selling merchandise	2,200.00 HKD	-	2,200.00 HKD	52,200.00 HKD	
01/09/2022	Re-used	Electronics	720p AHD Camera	89.04 HKD	8	712.32 HKD	51,487.68 HKD	
01/09/2022	Re-used	Electronics	Tether	1,200.00 HKD	1	1,200.00 HKD	50,287.68 HKD	
01/09/2022	Re-used	Electronics	CAN Transceiver (Beetle CJMCU)	33.39 HKD	12	400.68 HKD	49,887.00 HKD	
01/09/2022	Re-used	Electronics	CAN Transceiver (Beetle CAN PCB)	3.71 HKD	12	44.52 HKD	49,842.48 HKD	
01/09/2022	Re-used	Electronics	Optic Fibre	66.78 HKD	1	66.78 HKD	49,775.70 HKD	
01/09/2022	Re-used	Electronics	Motor Driver	5.57 HKD	2	11.14 HKD	49,764.56 HKD	
01/09/2022	Re-used	Electronics	Stepdown Converter(48V to 12V)	183.64 HKD	5	918.20 HKD	48,846.36 HKD	
01/09/2022	Re-used	Electronics	ESC	94.60 HKD	8	756.80 HKD	48,089.56 HKD	
01/09/2022	Re-used	Electronics	Raspberry Pi 4	1,000.00 HKD	1	1,000.00 HKD	47,089.56 HKD	
01/09/2022	Re-used	Hardware	Blue Robotics Thrusters	1,049.09 HKD	6	6,294.54 HKD	40,795.02 HKD	
01/09/2022	Re-used	Hardware	Power Supply	950.00 HKD	1	950.00 HKD	39,845.02 HKD	
01/09/2022	Re-used	Hardware	Gamepad (X-box Controller)	800.00 HKD	1	800.00 HKD	39,045.02 HKD	
01/09/2022	Re-used	Hardware	Anderson Powerpole	400.00 HKD	1	400.00 HKD	38,645.02 HKD	
01/09/2022	Re-used	Hardware	PU foam	200.00 HKD	-	200.00 HKD	38,445.02 HKD	
01/09/2022	Re-used	Hardware	Optic Fibre Receiver and Transmitter	779.05 HKD	1 set	779.05 HKD	37,665.97 HKD	
01/09/2022	Re-used	Hardware	AHD Capture card	555.35 HKD	1	555.35 HKD	37,110.62 HKD	
01/09/2022	Re-used	Hardware	Electromagnets	33.39 HKD	4	133.56 HKD	36,977.06 HKD	
01/09/2022	Re-used	Hardware	Laptop	13,599.23 HKD	1	13,599.23 HKD	23,377.83 HKD	
01/09/2022	Purchased	Hardware	Esun PLA+ Filaments	1,324.38 HKD	-	1,324.38 HKD	22,053.45 HKD	
01/09/2022	Purchased	Hardware	Waterproof Plug	7.94 HKD	60	476.40 HKD	21,577.05 HKD	
05/10/2022	Purchased	Electronics	Control Box Cables	22.37 HKD	20	447.40 HKD	21,129.65 HKD	
05/10/2022	Purchased	Electronics	22 inch Monitors	332.77 HKD	2	665.54 HKD	20,464.11 HKD	
10/11/2022	Purchased	Hardware	Aluminium Strips	34.00 HKD	4	136.00 HKD	20,328.11 HKD	
10/02/2023	Purchased	Hardware	D Hook	8.91 HKD	10	89.10 HKD	20,239.01 HKD	
13/02/2023	Purchased	Hardware	Gear Pump	44.52 HKD	1	44.52 HKD	20,194.49 HKD	
13/02/2023	Purchased	Electronics	BGC Electronics	166.94 HKD	1 set	166.94 HKD	20,027.55 HKD	
29/04/2023	Purchased	Travel	Acomodation in Hong Kong	8,000.00 HKD	-	8,000.00 HKD	12,027.55 HKD	
29/04/2023	Purchased	Travel	Transportation to Hong Kong	2,600.00 HKD	-	2,600.00 HKD	9,427.55 HKD	
-	Purchased	Travel	To Swimming Pool	1,457.10 HKD	-	1,457.10 HKD	7,970.45 HKD	
Total Raised:								52,200.00 HKD
Total Spent								44,229.55 HKD
Final Balance								7,970.45 HKD

Table 5: Project Costing Table (All currency in HKD)

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Appendix 1: Safety Procedures

Workshop Safety

Workshop Safety Checklist

General

- ☐ Safety equipment including safety glasses and protective gloves must be worn for using power tools and sharp tools
- ☐ Respirator must be worn when in contact with fine particles
- ☐ Clear work area to prevent hazards (knocking things over, tripping hazard, etc.)
- ☐ Sufficient amount of tools and equipment
- ☐ Tools and equipment are properly stored
- ☐ Regular sanitizing of work and personal space

Soldering

- ☐ Safety goggles and mask must be worn
- ☐ Ventilation must be turned on
- ☐ Work area must be clear

Handling Resin

- ☐ Ventilation must be turned on
- ☐ Prepare tissue paper
- ☐ Work area must be clear

Pneumatic

- ☐ Wear proper PPE
- ☐ Make sure no personnel is in the way of the pneumatic system's range of motion
- ☐ Make sure there is all the safety features on the pump, including: regulator, pressure relief valve, pressure release valve
- ☐ Make sure all personnel is aware of the operation of the pneumatic system
- ☐ The pressure of the pneumatic system should meet the MATE restrictions of 40PSI
- ☐ Make sure the wiring are secure
- ☐ Compressor must have no sign of any external rust

Operation Safety Checklist

Setup On Desk

- ☐ Area clear/safe (no tripping hazards, items in the way)
- ☐ Tether is laid out and managed by a team member
- ☐ Plugs and sockets are connected securely
- ☐ Verify power switch is off

- ☐ Thrusters are properly shielded
- ☐ No exposed copper or bare wires
- ☐ Screws and nuts are tight
- ☐ Tether securely connected to ROV
- ☐ Single inline 30A fuse in place

Power up

- ☐ Ensure all team members are attentive
- ☐ Call out, "Power on"

Power on

- ☐ Control computers up and running
- ☐ Call out, "Test thrusters"
- ☐ Perform thruster test
- ☐ Verify video feeds
- ☐ Test active manipulator

Launch

- ☐ Call out, "Prepare to launch"
- ☐ Deck crew members handling ROV call out, "Ready"
- ☐ Launch ROV

ROV Retrieval

- ☐ Pilot calls out, "ROV surfacing"
- ☐ Deck crew calls out, "ROV on surface", when ROV reaches the surface
- ☐ Stop thrusters
- ☐ Remove ROV from water

In Case Of Loss Of Communication

- ☐ Restart ROV
- ☐ Check status light
- ☐ Restart control program
- ☐ If communication restored, resume mission

Maintenance

- ☐ Verify thrusters are free of foreign objects and are spinning freely
- ☐ Visual inspection for any damage
- ☐ All cables are neatly secured
- ☐ Screws and nuts are tight